

Time series. Introduction

Modelització Estadística

Grau en Intel·ligència Artificial

Course 2025-26



Outline

- 1 Introduction to time series
- 2 Stationary series
- 3 ACF, PACF

Time series definition

Time series

Ordered sequence of observations of the same phenomenon. Typically measured at equally spaced successive instants of time.

$$\{X_t\}_{t=1,\dots,T} = \{X_1, X_2, \dots, X_T\}$$

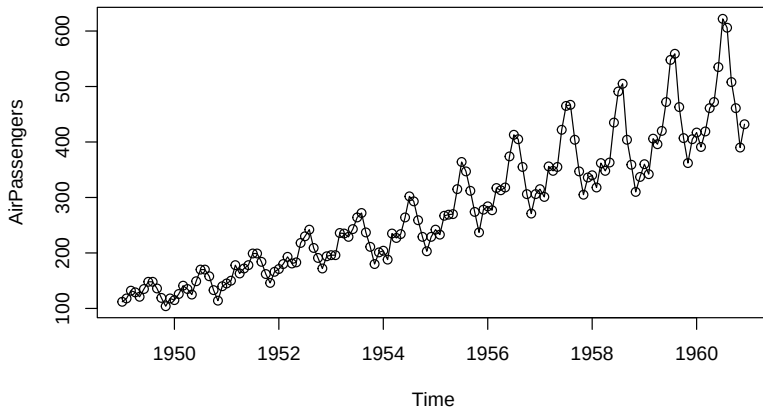
Example

AirPassengers: Monthly totals of international airline passengers in USA, 1949 to 1960 (Box & Jenkins, 1976)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1949	112	115	145	171	196	204	242	284	315	340	360	417
1950	118	126	150	180	196	188	233	277	301	318	342	391
1951	132	141	178	193	236	235	267	317	356	362	406	419
1952	129	135	163	181	235	227	269	313	348	348	396	461
1953	121	125	172	183	229	234	270	318	355	363	420	472
1954	135	149	178	218	243	264	315	374	422	435	472	535
1955	148	170	199	230	264	302	364	413	465	491	548	622
1956	148	170	199	242	272	293	347	405	467	505	559	606
1957	136	158	184	209	237	259	312	355	404	404	463	508
1958	119	133	162	191	211	229	274	306	347	359	407	461
1959	104	114	146	172	180	203	237	271	305	310	362	390
1960	118	140	166	194	201	229	278	306	336	337	405	432

Time series definition

```
plot(AirPassengers, type="o")
```



Motivation and Objectives

Motivation

Describing and forecasting time series is crucial in different areas of knowledge:

- Epidemiology
- Finance
- Econometrics
- Signal processing

Objectives

- **Description**: Describe temporal patterns in a time series: regular and/or seasonal effects, cyclicity, trends, outliers, sudden changes, breaks, ...
- **Estimation**: Estimate the values of the time series parameters.
- **Validation**: Validate if the estimated model is correct or not.
- **Prediction/Forecasting**: Predict future values of the time series.

Exploratory Data Analysis

Plot of the series and identification of the components:

- **Trend (T_t)**: Long term tendency

- Moving average of order s :

$$T_t = \frac{1}{s} \sum_{i=1}^s X_{t-s/2+i}$$

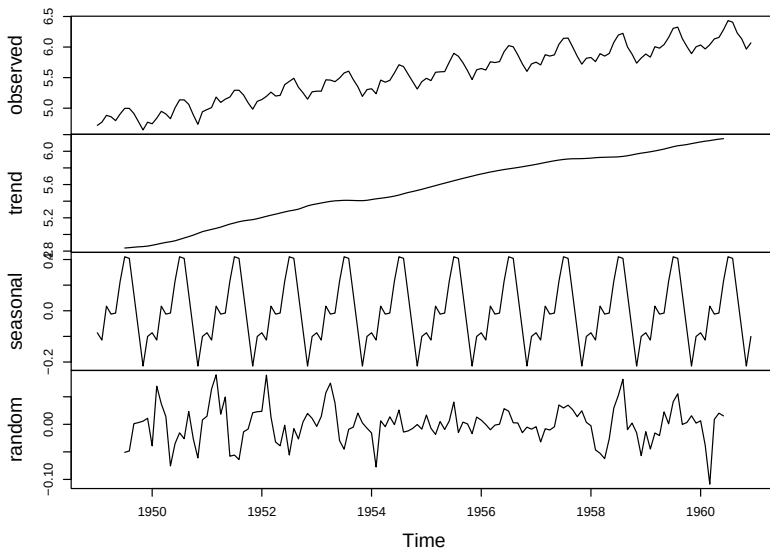
- **Seasonal (S_t)**: Pattern repeated periodically with the same period
 - Seasonal index: Mean for each period of detrended series ($X_t - T_t$)
- **Cycle (C_t)**: Pattern repeated periodically with non-constant period
 - Not easy to model due to the changing period.
- **Random (w_t)**: Random noise
 - Remainder part: $w_t = X_t - T_t - S_t - C_t$

Additive model:

$$X_t = T_t + S_t + C_t + w_t$$

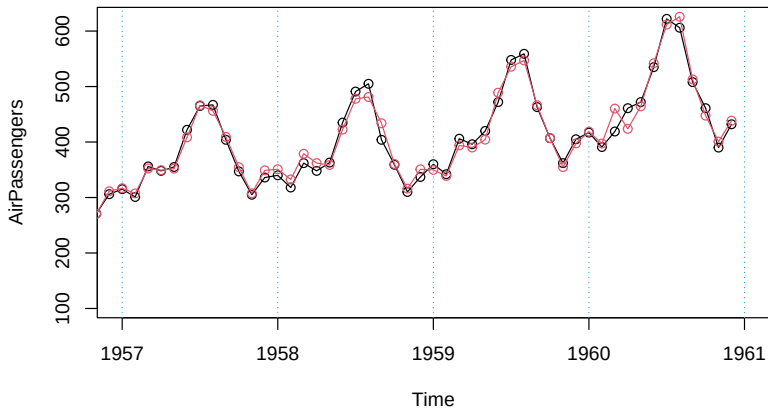
Time series Decomposition

Decomposition of additive time series



Time Series Modelling

Goal: Find a mathematical model that reflects the behaviour of the observed data



Time Series Modelling

Deterministic model

The expected value of X_t depends on a parametric function F of t and the random component does not depend on the previous values.

$$X_t = F(t) + Z_t \quad Z_t \sim N(0, \sigma_Z^2)$$

Stochastic Model

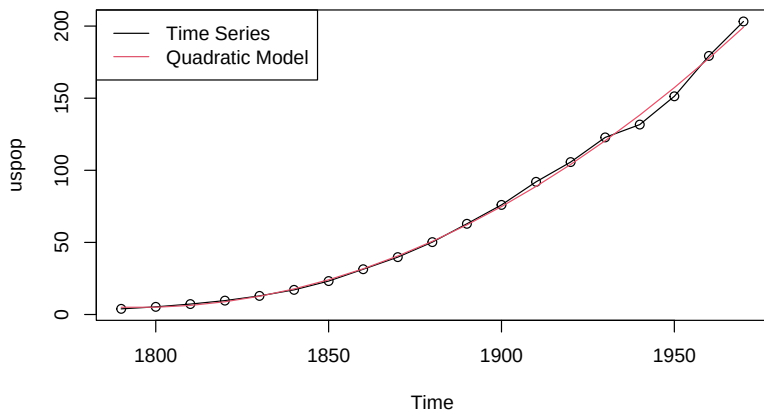
The expected value of X_t depends on the previous values $X_{t-1}, X_{t-2}, \dots$ and/or the previous random components $Z_{t-1}, Z_{t-2}, \dots$ plus a random component independent of the past.

$$X_t = G(X_{t-1}, X_{t-2}, \dots, Z_{t-1}, Z_{t-2}, \dots) + Z_t \quad Z_t \sim N(0, \sigma_Z^2)$$

Time Series Examples (1/10)

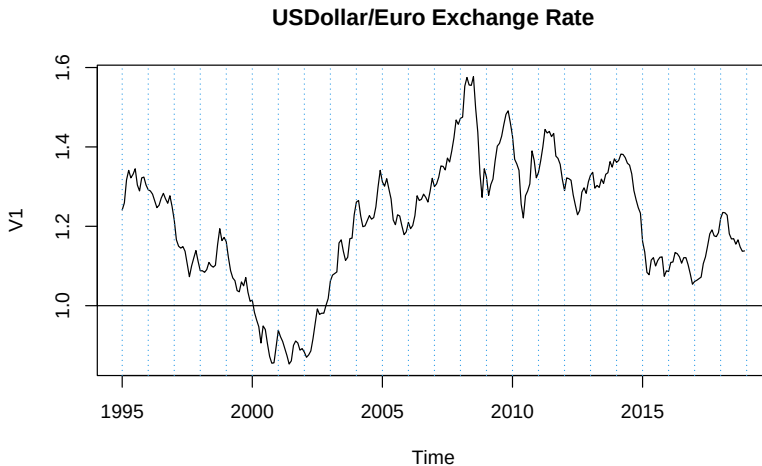
Example 1.1: Population recorded by US Census, 19 decades, 1790 to 1970.

US Census population



Time Series Examples (2/10)

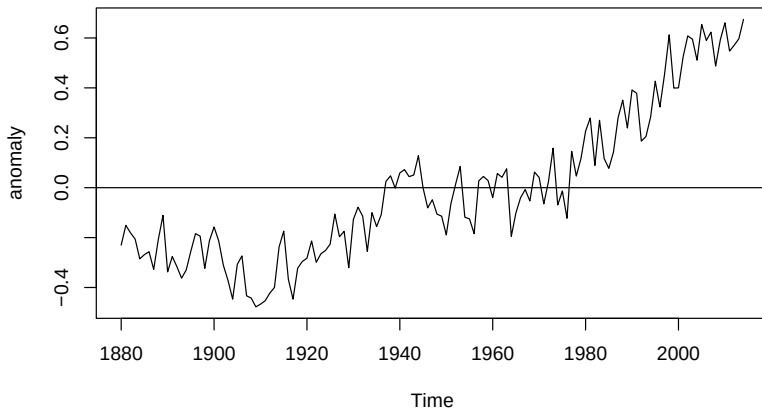
Example 1.2: Exchange Rate Dollar/Euro (ECU before 1999). Monthly mean.
Source: Bank of Spain



Time Series Examples (3/10)

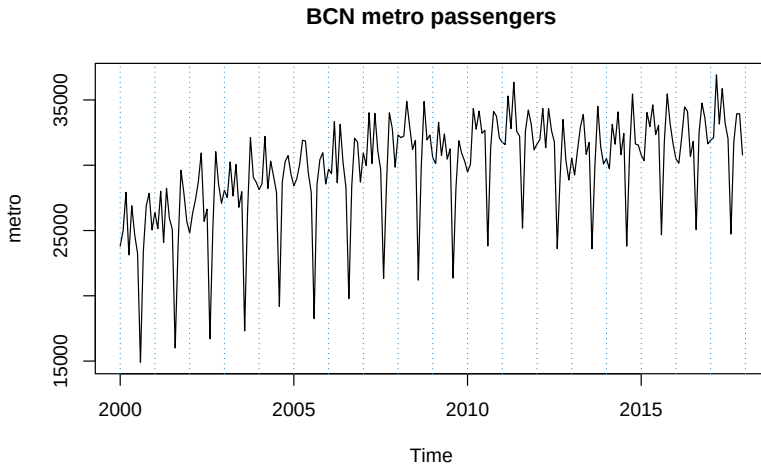
Example 1.3: Global Warming: Yearly average global temperature deviations (1880-2009) in degrees centigrade. Source: NASA

Global Warming



Time Series Examples (4/10)

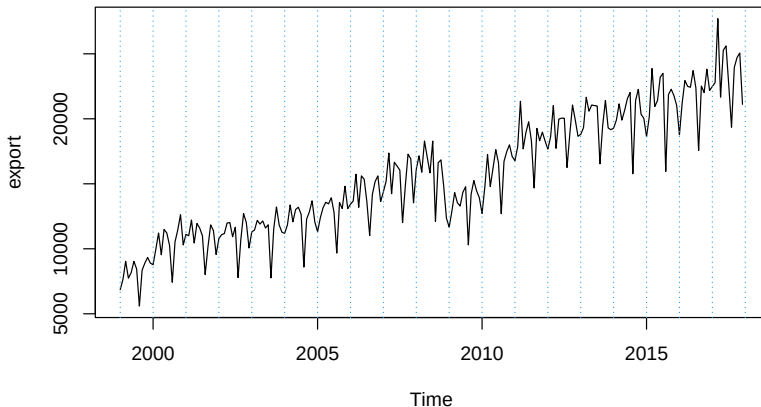
Example 1.4: Barcelona metro passengers (thousands). Monthly data. Source: INE



Time Series Examples (5/10)

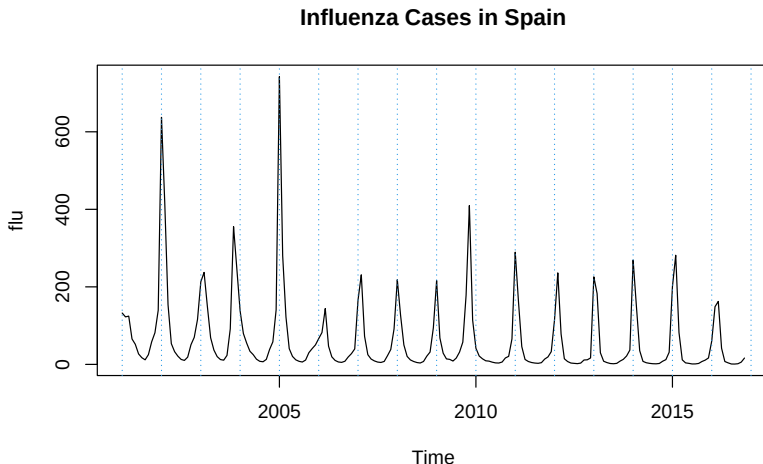
Example 1.5: Spain: Total Exports (thousand of millions). Source: Ministry of industry, trade and tourism of Spain

Spain Exports



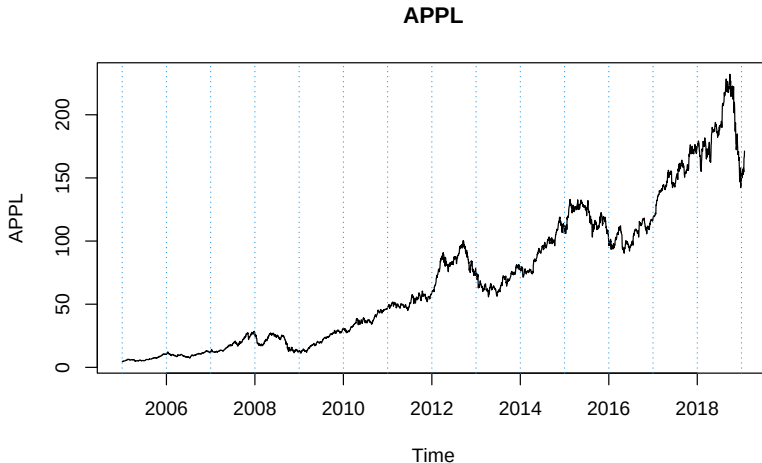
Time Series Examples (6/10)

Example 1.6: Number of reported cases of influenza affected (thousand). Monthly data. Source: Ministry of Health of Spain



Time Series Examples (7/10)

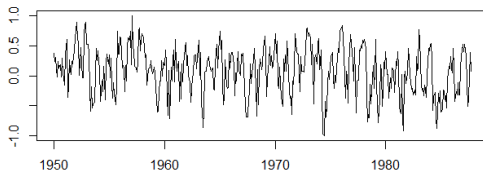
Example 1.7: Apple Inc.(AAPL) NasdaqGS Real Time Price. Currency in USD



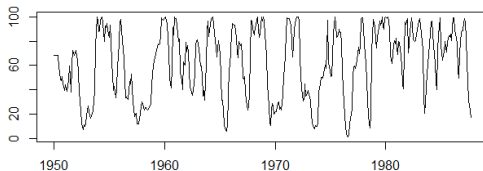
Time Series Examples (8/10)

Example 1.8: El Niño and Fish Population. Monthly Southern Oscillation Index (SOI) and Recruitment (estimated new fish), 1950-1987.

Southern Oscillation Index



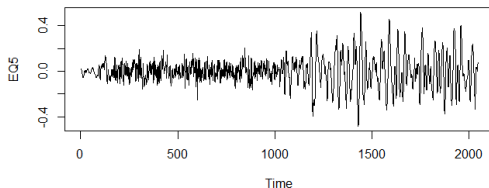
Recruitment



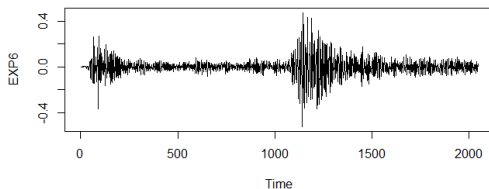
Time Series Examples (9/10)

Example 1.9: Earthquakes and Explosions (Arrival phases from an earthquake (top) and explosion (bottom) at 40 points per second.

Earthquake

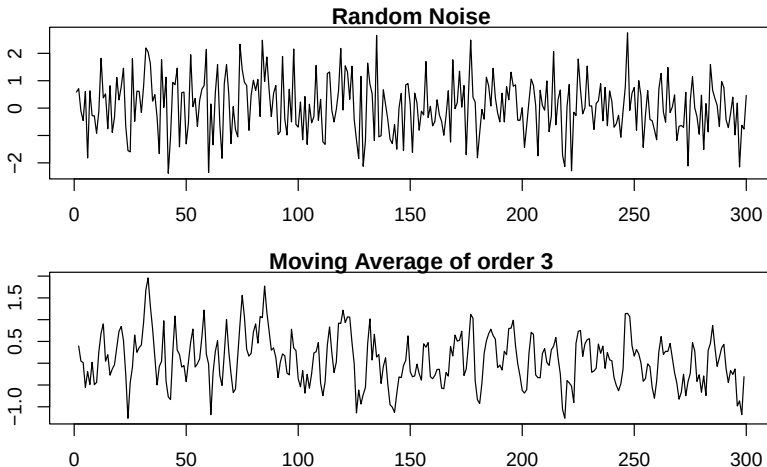


Explosion

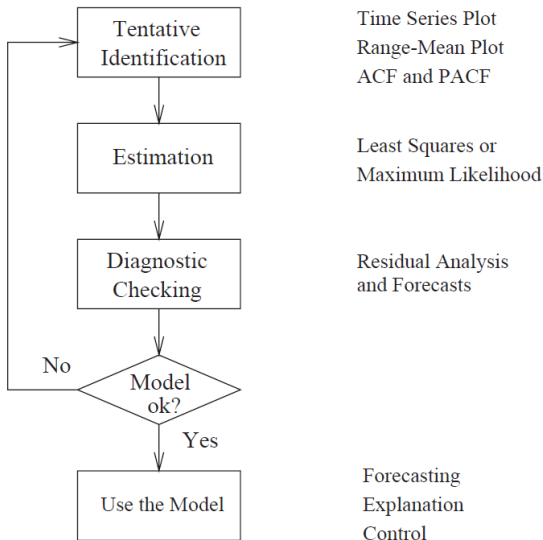


Time Series Examples (10/10)

Example 1.10: Gaussian white noise series (top) and three-point moving average of the Gaussian white noise series (bottom).



Box-Jenkins Methodology



Distribution of a general stochastic process

- **First and second moments** for the multivariate distribution of $\{X_t\}_{t=1..T}$

$$E[(X_1, X_2, \dots, X_T)] = (\mu_1, \mu_2, \dots, \mu_T)$$

$$\text{Var}((X_1, X_2, \dots, X_T)) = \begin{pmatrix} \sigma_1^2 & \sigma_{1,2} & \sigma_{1,3} & \dots & \sigma_{1,T} \\ \sigma_{1,2} & \sigma_2^2 & \sigma_{2,3} & \dots & \sigma_{2,T} \\ \sigma_{1,3} & \sigma_{2,3} & \sigma_3^2 & \dots & \sigma_{3,T} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_{1,T} & \sigma_{2,T} & \sigma_{3,T} & \dots & \sigma_T^2 \end{pmatrix}$$

- **Parameters of the model**

- T values for the mean: $E(X_t) = \mu_t$
- T values for the variances: $V(X_t) = \sigma_t^2$
- $T * (T - 1)$ values for the covariances: $\text{Cov}(X_t, X_s) = \sigma_{t,s}$

Distribution of an stationary stochastic process

- **First and second moments** for the multivariate distribution of $\{X_t\}_{t=1..T}$

$$E[(X_1, X_2, \dots, X_T)] = (\mu, \mu, \dots, \mu)$$

$$\text{Var}((X_1, X_2, \dots, X_T)) = \begin{pmatrix} \sigma^2 & \sigma_1 & \sigma_2 & \dots & \sigma_{T-1} \\ \sigma_1 & \sigma^2 & \sigma_1 & \dots & \sigma_{T-2} \\ \sigma_2 & \sigma_1 & \sigma^2 & \dots & \sigma_{T-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_{T-1} & \sigma_{T-2} & \sigma_{T-3} & \dots & \sigma^2 \end{pmatrix}$$

- **Parameters of the model**

- 1 value for the mean: $E(X_t) = \mu$
- 1 value for the variances: $V(X_t) = \sigma^2$
- $T - 1$ values for the covariances: $\text{Cov}(X_t, X_s) = \sigma_{|t-s|}$

Stationary Series

Strict Stationary processor series

The joint distribution of the whole series does not depend on the time origin

$$F_{(X_1, \dots, X_t)}(x_1, \dots, x_t) = F_{(X_{1+s}, \dots, X_{t+s})}(x_{1+s}, \dots, x_{t+s}) \quad \forall t, s$$

Weakly Stationary process or series

The two first moments of the multivariate distribution of the whole series does not depend on the time origin:

- constant mean (μ)
- constant variance (σ^2)
- constant autocovariance structure ($\sigma_{t,s} = \sigma_{|t-s|}$)

The latter refers to the covariance between X_t and X_{t-1} being the same as X_{t-s} and X_{t-s-1} .

Property

Weakly Stationary Process + Gaussian multivariate Distribution
 $\Rightarrow$ **Strict Stationary Process**

Stationary Series

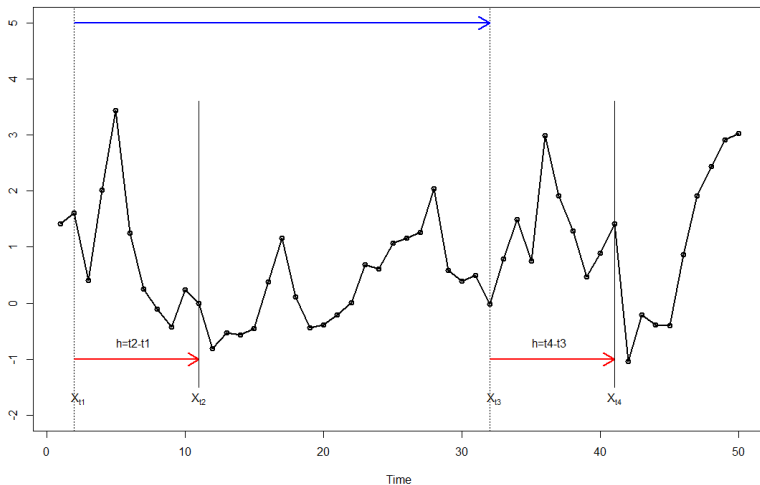


Figure 1: Example of an stationary process

Stationary Series

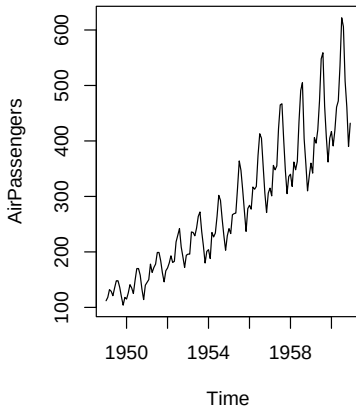
- Is our data stationary?
- How can we detect?
- In general:
 - **Plot** the data
 - Identify non stationary components (trends, seasonal patterns, cycles)
 - Transform the series to remove those components
 - For the transformed (stationary) series, plot and analyze the sample autocorrelation

Transformations: Change the Scale

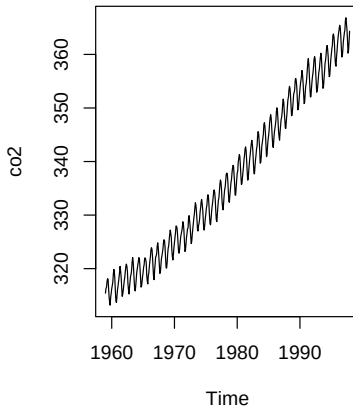
Is the variance constant?

It is very common that the variance of the series increases when the level of the series rises:

Non Constant Variance



Constant Variance



Transformations: Change the Scale

Tools to diagnose the non-constant variance:

1 Mean-Variance plot:

- Calculate the mean and the variance of consecutive groups of 8-12 observations
- Plot the variance against the mean of each group

2 Boxplot for periods:

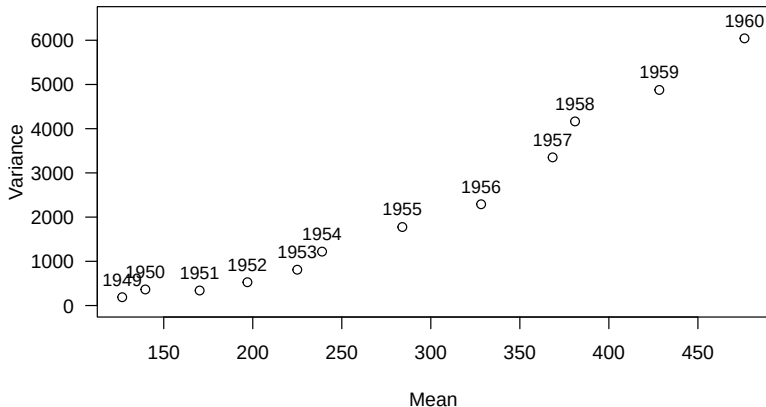
- Represent the boxplot for each group of 8-12 observations
- The height of the box (IQR) is a robust estimate of variability
- If the variance is similar for all the groups $\Rightarrow$ No scale transformation
- If the variance is higher for higher values of the mean $\Rightarrow$ Change the scale
 - Box-Cox transformation:

$$\begin{cases} \frac{X^\lambda - 1}{\lambda} & \lambda \in [-1, 2], \lambda \neq 0 \\ \log(X) & \lambda = 0 \end{cases}$$

- **Note:** Usually the log transformation is applied (easy to interpret)

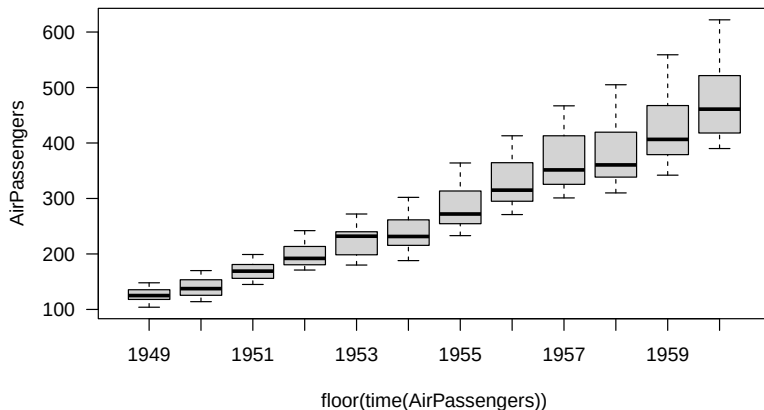
Transformations: Change the Scale

```
m <- by(AirPassengers, floor(time(AirPassengers)), mean)
v <- by(AirPassengers, floor(time(AirPassengers)), var)
plot(v~m, xlab="Mean", ylab="Variance",ylim=c(0,6500),las=1)
text(m,v,1949:1960,cex=0.9, adj=0.5, pos=3)
```



Transformations: Change the Scale

```
boxplot(AirPassengers~floor(time(AirPassengers)),las=1)
```



Transformation: Seasonal difference

Is there a Seasonal Pattern?

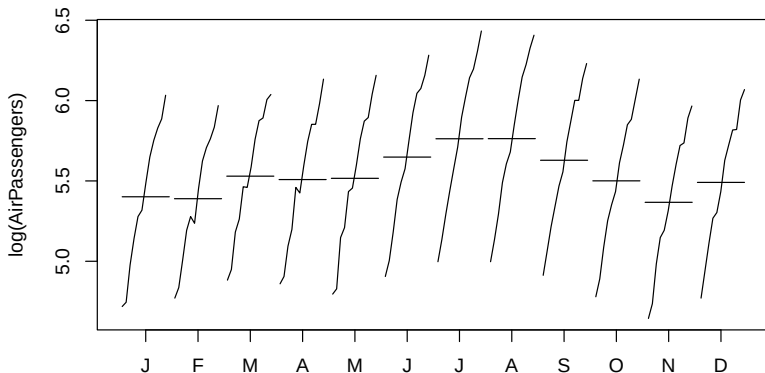
- A similar pattern for a constant period s is observed
 - Monthly data: $s=12$ observations
 - Quarterly data: $s=4$ observations
 - Daily data: $s=7$ observations
 - Hourly data: $s=24$ observations
- This pattern is the so-called **Seasonal Pattern** of the time series.
- To remove this pattern, a linear filter is applied to the series
 - Moving Average of order s : $W_t = \frac{1}{s} \sum_{i=1}^s X_{t-i+1}$
 - Seasonal difference of order s : $W_t = X_t - X_{t-s} \quad t > s$

Note: The seasonal difference is preferred and includes the other option

Transformation: Seasonal difference

Tool to diagnose the seasonal pattern:

```
monthplot(log(AirPassengers))
```



Transformation: Seasonal difference

Notation

Backshift operator

$$BX_t = X_{t-1} \qquad B^s X_t = X_{t-s}$$

(same as lag operator **L** in some articles/books)

- Moving Average of order s :

$$W_t = \frac{1}{s} \sum_{i=1}^s X_{t-i+1} = (1 + B + \dots + B^{s-1})X_t$$

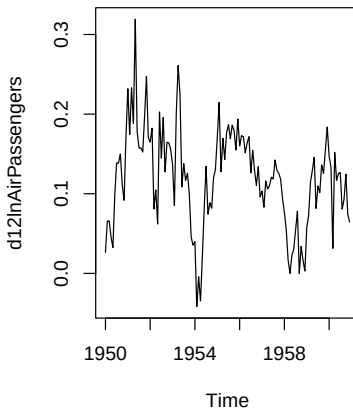
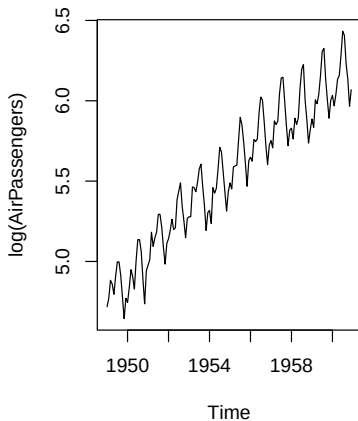
- Seasonal Difference of order s :

$$W_t = X_t - X_{t-s} = (1 - B^s)X_t = \nabla_s X_t$$

Note: The seasonal difference is equivalent to a regular difference of a moving average of order s .

$$(1 - B^s) = (1 - B)(1 + B + \dots + B^{s-1})$$

Transformation: Seasonal difference



Transformation: Regular difference

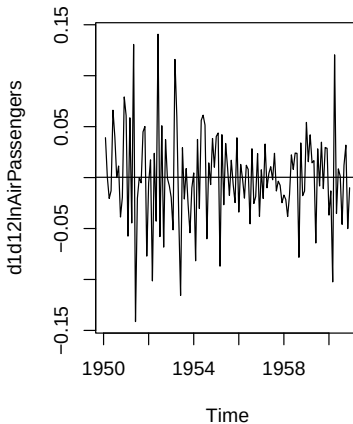
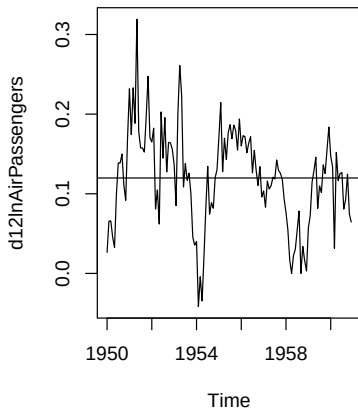
Is the mean constant?

- Linear or general trend implies non constant mean.
- A regular difference is applied until the mean can be considered constant

$$W_t = X_t - X_{t-1} = (1 - B)X_t$$

- **Overdiferentation:** If the differenced time series yields a higher variance, then the later difference is not needed

Transformation: Regular difference



Transformation into stationary time series

NON-STATIONARITY CAUSE	TRANSFORMATION
Non-constant Variance	Box-Cox Transformation ($\lambda \in [-1, 2]$) $W_t = \frac{x_t^\lambda - 1}{\lambda} \quad \lambda \neq 0$ $W_t = \log X_t \quad \lambda = 0$
Linear Deterministic Trend Non-constant mean	Regular difference $W_t = (1 - B)X_t$ Regular difference $W_t = (1 - B)X_t$
Deterministic d-order polynomial Trend Stochastic Trend	d-Regular differences $W_t = (1 - B)^d X_t$ d-Regular differences $W_t = (1 - B)^d X_t$ until stationary W_t
Seasonal pattern of order s	Seasonal difference $W_t = (1 - B^s)X_t$
Indexes and Financial data	log>Returns: $W_t = (1 - B) \log X_t \cong \frac{X_t - X_{t-1}}{X_{t-1}}$

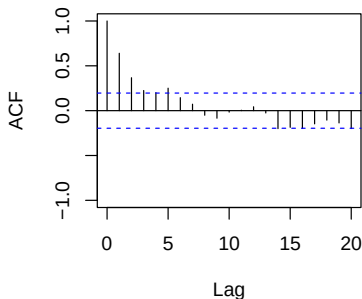
ACF, PACF: Moments of Stationary Processes

MOMENT	THEORETICAL	SAMPLE
Mean	μ	$\bar{X}_t = \frac{1}{T} \sum_{t=1}^T X_t$
Autocovariance $\{\gamma(k)\}$	$E[(X_{t+k} - \mu)(X_t - \mu)]$	$\frac{1}{T} \sum_{t=1}^{T-k} (X_{t+k} - \bar{X})(X_t - \bar{X})$
Variance $\{\sigma_X^2 = \gamma(0)\}$	$E[(X_t - \mu)^2]$	$\frac{1}{T} \sum_{t=1}^T (X_t - \bar{X})^2$
Autocorrelation $\{\rho(k) = \gamma(k)/\gamma(0)\}$	$\frac{E[(X_{t+k} - \mu)(X_t - \mu)]}{E[(X_t - \mu)^2]}$	$\frac{\sum_{t=1}^{T-k} (X_{t+k} - \bar{X})(X_t - \bar{X})}{\sum_{t=1}^T (X_t - \bar{X})^2}$

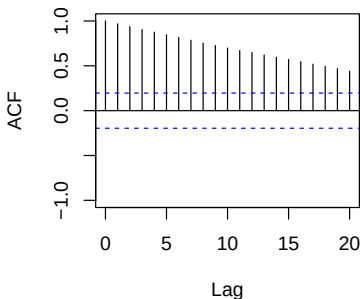
ACF, PACF: Correlogram

- **Autocorrelation Function (ACF)**: measures the relationship between the two k -lag apart variables, X_t and X_{t+k} .
- **ACF** lies between -1 and +1
- Correlogram is the plot of the ACF $\rho(k)$ against k
- Under **Stationarity**: ACF falls immediately from 1 to 0
- Under **Non-stationary**: the ACF declines gradually from 1 to 0 over a prolonged period of time

Stationary



Non-Stationary



ACF, PACF: Correlogram

Variance of the sample ACF:

- For large sample size T , asymptotically:

$$V(\hat{\rho}(k)) \approx \frac{1}{T}$$

The sample ACF represents the values of $\hat{\rho}(k)$ for each lag k from $k = 1, 2, \dots$. The confidence bands are calculated using the asymptotic distribution for the estimator:

$$\pm \frac{1.96}{\sqrt{T}}$$

For each lag k we can test its significance by using the plot:

- If $\hat{\rho}(k)$ lies between the confidence bands, we cannot reject the null hypothesis ($H_0 : \rho(k) = 0$) and the theoretic autocorrelation for this lag can be considered null.

ACF, PACF: Partial ACF

PACF: Partial correlation (of a stationary process) is the relationship between two variables, after excluding the effect of one or more independent variables.

In other words:

- $\phi_{11} = \text{cor}(X_{t+1}, X_t) = \rho(1)$
- $\phi_{hh} = \text{cor}(X_{t+h} - \hat{X}_{t+h}, X_t - \hat{X}_t), h \geq 2$

Example

In a regression context in which y represents the response variable and x_1 , x_2 , and x_3 are predictor variables, the **partial correlation** between y and x_3 is the correlation between those variables but taking into account how both y and x_3 are related to x_1 and x_2

ACF, PACF: Partial ACF

Partial Autocorrelation function

Estimation by *Ordinary Least Squares*:

$$x_t = \phi_{1,1}x_{t-1} + Z_t$$

$$x_t = \phi_{1,2}x_{t-1} + \phi_{2,2}x_{t-2} + Z_t$$

$$x_t = \phi_{1,3}x_{t-1} + \phi_{2,3}x_{t-2} + \phi_{3,3}x_{t-3} + Z_t$$

:

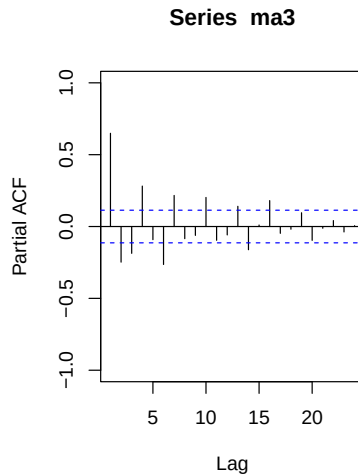
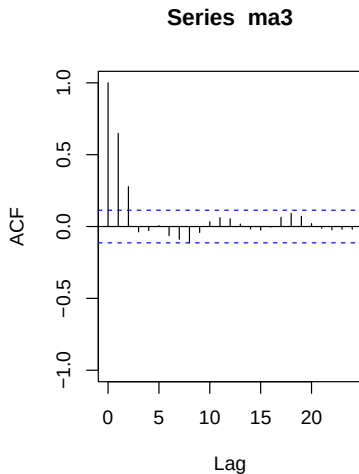
$$x_t = \phi_{1,h}x_{t-1} + \phi_{2,h}x_{t-2} + \phi_{3,h}x_{t-3} + \dots + \phi_{h,h}x_{t-h} + Z_t$$

:

PACF: $\{\phi_{1,1}, \phi_{2,2}, \dots, \phi_{h,h}, \dots\}$

ACF, PACF: Estimation of Correlation

Sample ACFs and PACF



{Standard R: Sample ACF begins at 0 but sample PACF begins at 1}

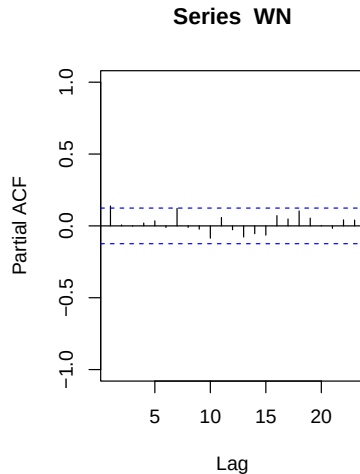
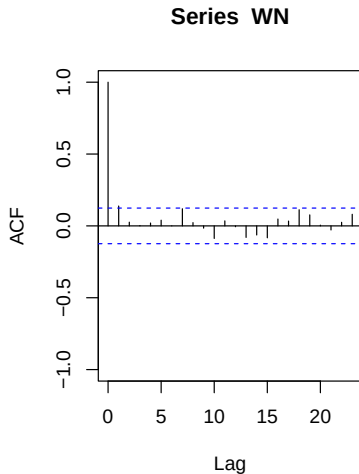
*Autocorrelation of white noise (Z_t)

$$Z_t \sim WN(\sigma_Z^2) \sim N(0, \sigma_Z^2) \text{ independent}$$

MOMENT	THEORETICAL
Mean	0
Autocovariance $\{\gamma(k)\}$	0
Variance $\{\gamma(0)\}$	σ_Z^2
Autocorrelation $\{\rho(k) = \gamma(k)/\gamma(0)\}$	0

ACF, PACF: Estimation of Correlation

Sample **ACF** and **PACF** for a white noise series



Bibliography

Textbook

- Shumway R., Stoffer D. (2016). *Time Series Analysis and Its Applications - With R Examples*.

References

- Box G., Jenkins G., Reinsel G. (2008). *Time series Analysis: forecasting and control*
- Peña D. (2005). *Análisis de Series Temporales*